Quantitative model
Overnight review

12 September 2026, 13:06 UTC

The first fertility window now matches. The complete historical transition and new policies are not yet verified.

Initial calibration: loss 158.541191, down 0.438% from 159.238986. Three bounded batches completed 252 new trials; the final winner reproduced twice. Annual beta remains estimated at its 0.99 cap. This is a modest refinement, not a resolved fit.

First historical window: the six-date forecast with a permanent preference decline of 0.01414 produces 1.974856 against 1.974875. The absolute error is 0.00001945. Housing and pension residuals pass their unchanged gates. This is a short-horizon result, not a four-window historical fit.

Long forecast: the earlier, larger decline of 0.045 also cleared the 28-date housing/PAYGO root. It failed terminal-distance checks: terminal household mass remained 12.56% from the stationary endpoint and the pension benefit 7.78% away. A finite equilibrium is not sufficient for horizon certification.

Numerical diagnosis: changing only the terminal root starting guesses allowed smaller preference declines to solve. Earlier failures at the initial guess did not prove equilibrium nonexistence. Separate age-advancement mass failures remain unresolved; their tolerance was not relaxed.

Still missing: an accepted sequence of four unexpected shocks through 2023, horizon verification of that sequence, and the new property-tax/rebate policy comparisons. No new policy effect should enter slides as an established result.

Current follow-up: independent 28- and 56-date checks of the preference that matches the first short window. They use verified numerical starts and unchanged source/target/fiscal contracts. The initial calibration candidate is not silently swapped beneath a historical run.

Reading order: page 2 shows every target; page 3 every parameter/restriction; page 4 the actual forecast shapes; page 5 the interpretation limits. The appendix preserves the complete standard initial-calibration graph set.

Complete initial target fit

All 12 scored moments plus the separate 2.1 normalization. Weight means the actual objective weight; the normalization has no loss contribution. Labels and values are read directly from the collected table.

Moment	Target	Model	Gap	Weight	Loss
Initial model completed fertility	2.1	2.1	8.9827e-07	-	-
Childless women, ages 40–44	0.198279	0.206956	0.00867741	35532.3	2.67549
Exactly one child among mothers, ages 40–44	0.213655	0.20658	-0.00707496	26952.8	1.34912
Period mean first-birth age	25.9763	26.1032	0.126914	139.828	2.25224
First births at age 30+	0.249278	0.232027	-0.0172509	13866.1	4.12646
Wealth / annual gross labor earnings	6.14586	4.84892	-1.29695	7.5951	12.7755
Annual bequests / aggregate wealth	0.0088	0.00858255	-0.000217451	5.16529e+06	0.24424
Old wealth/income p90 / median, ages 76–84	3.51594	4.43839	0.922459	10.6164	9.03378
Mean occupied rooms, capped at 9	5.5611	6.29059	0.729492	128.021	68.1272
Ownership, heads 30–55	0.648334	0.52444	-0.123894	2339.36	35.9088
First-birth room response, −1 to +3	0.720246	1.01103	0.29078	137.565	11.6315
Rooms: 3+ versus 1–2 resident children (model dependent proxy)	0.347067	0.200474	-0.146593	280.528	6.02841
Recent-parent ownership gap	0.162896	0.15016	-0.0127356	27055.8	4.38837

Sum of the 12 scored contributions: **158.541191063**. Complete empirical definitions, uncertainty and provenance are retained in selected_target_fit.csv. The first-birth rooms target remains 0.7202462623815278 with its original weight. No target was dropped or reweighted.

Parameters and restrictions

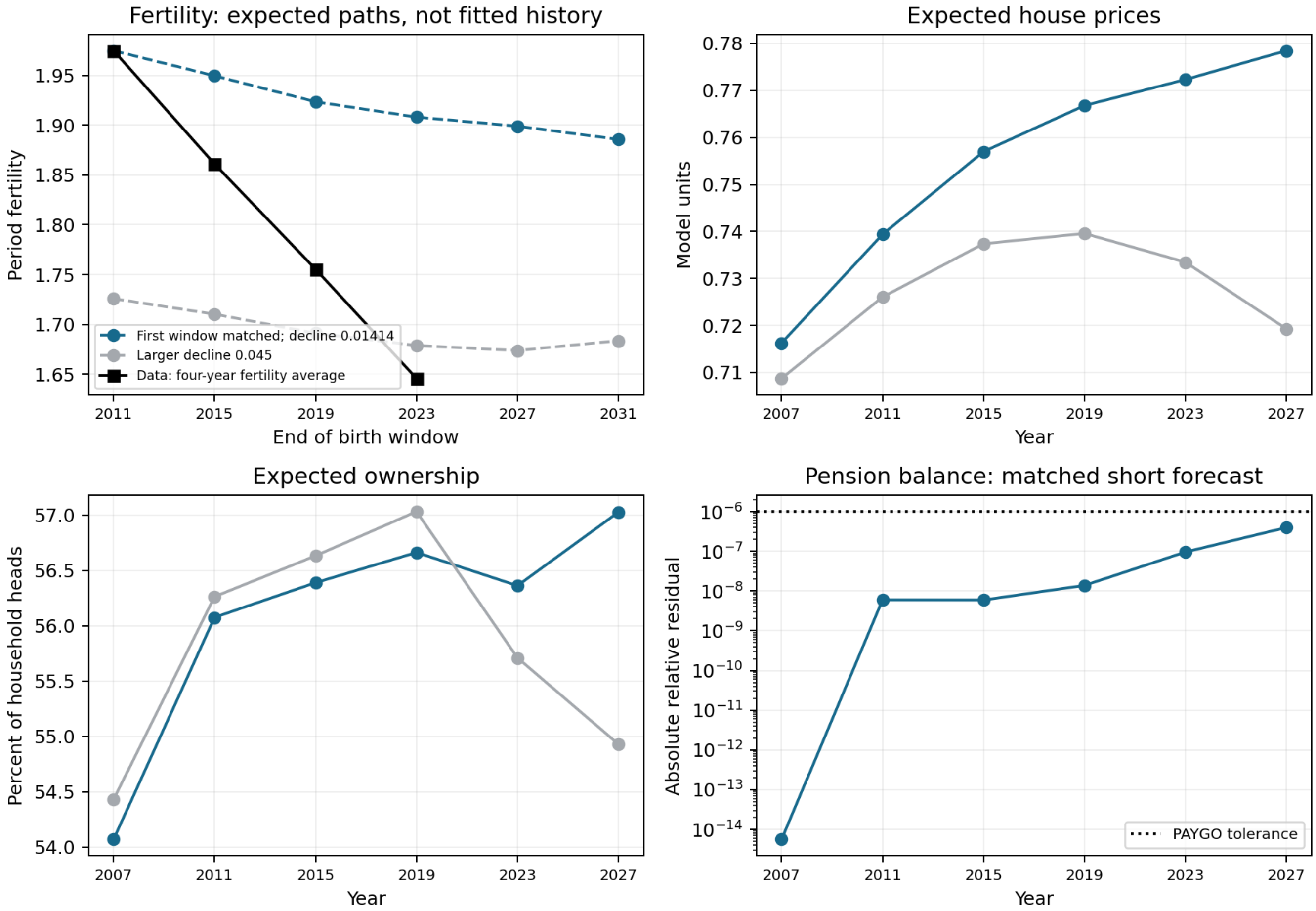
Nine structural coordinates remain free. Beta is estimated over [0.94, 0.99], not fixed. The table preserves the search bounds and near-bound flags; derived or external restrictions have no search bounds.

Parameter	Estimate	Lower	Upper	Near bound	Status
beta_annual	0.99	0.94	0.99	True	estimated_capped
kappa_fert	0.279392	0.02	50	True	diagnostic candidate; not a certified estimate
kappa_fert_continuation	0.346867	0.02	50	True	diagnostic candidate; not a certified estimate
chi	1.04803	0.1	5	False	diagnostic candidate; not a certified estimate
H0	8.19051	0.2	80	False	diagnostic candidate; not a certified estimate
theta0	0.0869545	0	8	False	diagnostic candidate; not a certified estimate
theta1	0.0571137	0.02	16	True	diagnostic candidate; not a certified estimate
first_birth_fixed_cost	0.1268	0	8	False	diagnostic candidate; not a certified estimate
h_P	2.3	0.1	2.3	True	diagnostic candidate; not a certified estimate
hbar_child_rooms	0	-	-	False	zero restriction
psi_child	0.159621	-	-	False	normalized to 2.1
payroll_tax	0.179	-	-	False	externally fixed
pension_period	2.04636	-	-	False	budget derived
housing_supply_elasticity	0.63	-	-	False	externally fixed
tenure_choice_kappa	0.005	-	-	False	externally fixed
alpha_cons	0.733	-	-	False	externally fixed
sigma	2	-	-	False	externally fixed

The stationary pension benefit 2.046361 is budget-derived. Its actual payroll revenue is 0.530413855, pension outlays 0.530413857, and relative gap -3.74e-09. The historical bridge changes age weights; its balanced pension need not equal this stationary benefit.

What the short forecasts actually show

Every model line is the forecast made after one permanent preference change. Future surprises are absent. Matching the first point does not match the later data. Neither six-date line passes the terminal-horizon checks.



Matched-window native root: maximum housing residual 3.2225e-05 (tolerance 0.0002); maximum PAYGO residual 3.97512e-07 (tolerance 0.000001). The full root receipt and dated source rows are saved with the figure.

The grey curve uses the larger decline that was extended to 28 dates. The blue curve uses the smaller decline that matches the first window and is now being checked at longer horizons. These are different preference experiments under the same pinned historical initial condition.

Interpretation and remaining checks

Preference change	First-window fertility	Gap to data	Short root	Horizon
-0.01414	1.974856	-0.000019	Pass	Not certified
-0.0225	1.907530	-0.067345	Pass	Not certified
-0.0275	1.867262	-0.107613	Pass	Not certified

Information: at each shock date households learn the current preference and expect it to persist. They do not anticipate later preference surprises. A full history must carry the actual household distribution and birth queues from each implemented first period.

Measurement: the fitted flow uses the retained household-rate analogue of female TFR. The 2.1 stationary normalization is a different object. Published TFR targets are four-year averages for windows ending 2011, 2015, 2019 and 2023. This measurement approximation remains explicit.

Population: historical head-age totals are imposed through the existing bridge and the 2023 person anchor. They are not independent model predictions. The outside-origin entry share 0.169 remains a diagnostic/outstanding closure object, not an estimated production normalization.

Fiscal closure: payroll revenue and pension outlays must balance at every accepted date. Property-tax rebates require a separate budget equation. A verified stationary or short-path fiscal balance is not evidence that all policy paths are solved.

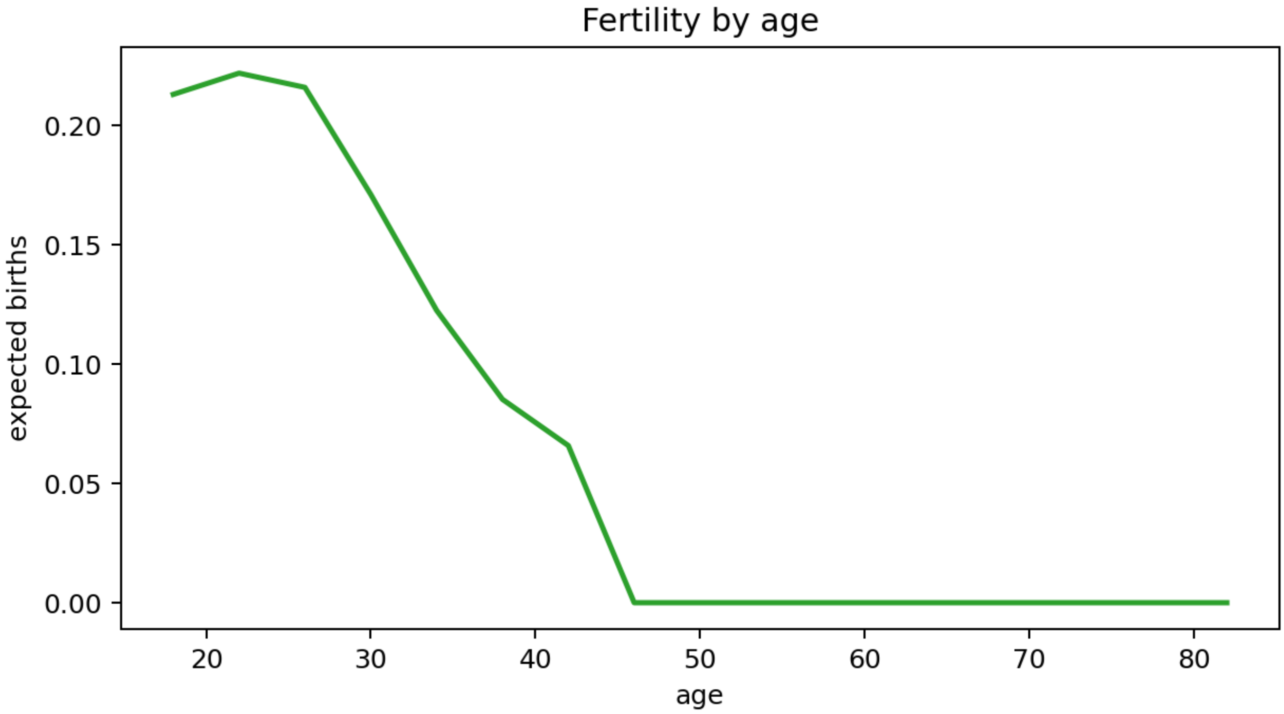
Calibration provenance: the final overnight initial refinement and the historical workers use distinct, explicitly pinned candidates. Their losses and parameter tables must not be silently combined. The historical workers retain the previously repeated capped candidate; the new initial refinement is a separate result.

What deserves immediate attention: whether the first-window match survives a longer horizon; then fitting the remaining unexpected shocks; then running the tax/rebate comparisons on that accepted inherited 2023 state. The unresolved mass-gate failures and measurement/entry conventions remain visible.

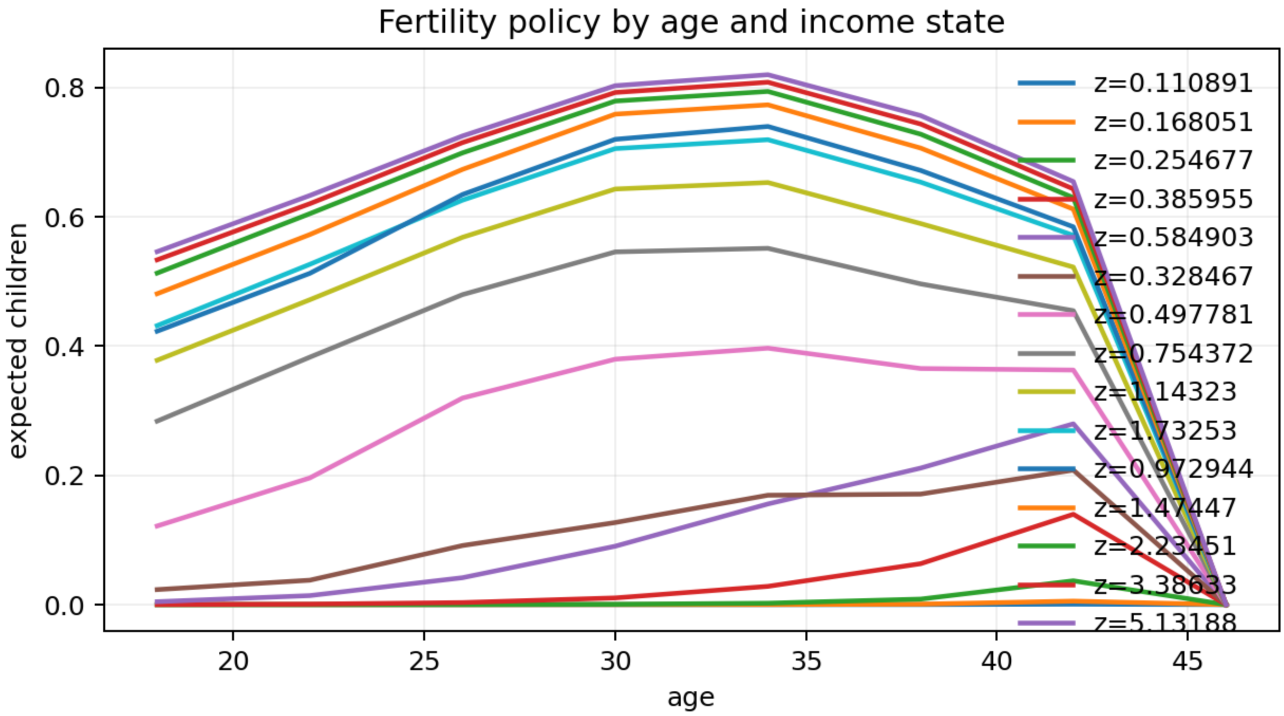
Standard initial-calibration diagnostics

Complete unchanged graph set. These are stationary candidate diagnostics, not evidence of a fitted 2007-2023 path.

fertility_by_age



fertility_policy_by_age_income_state



Standard initial-calibration diagnostics

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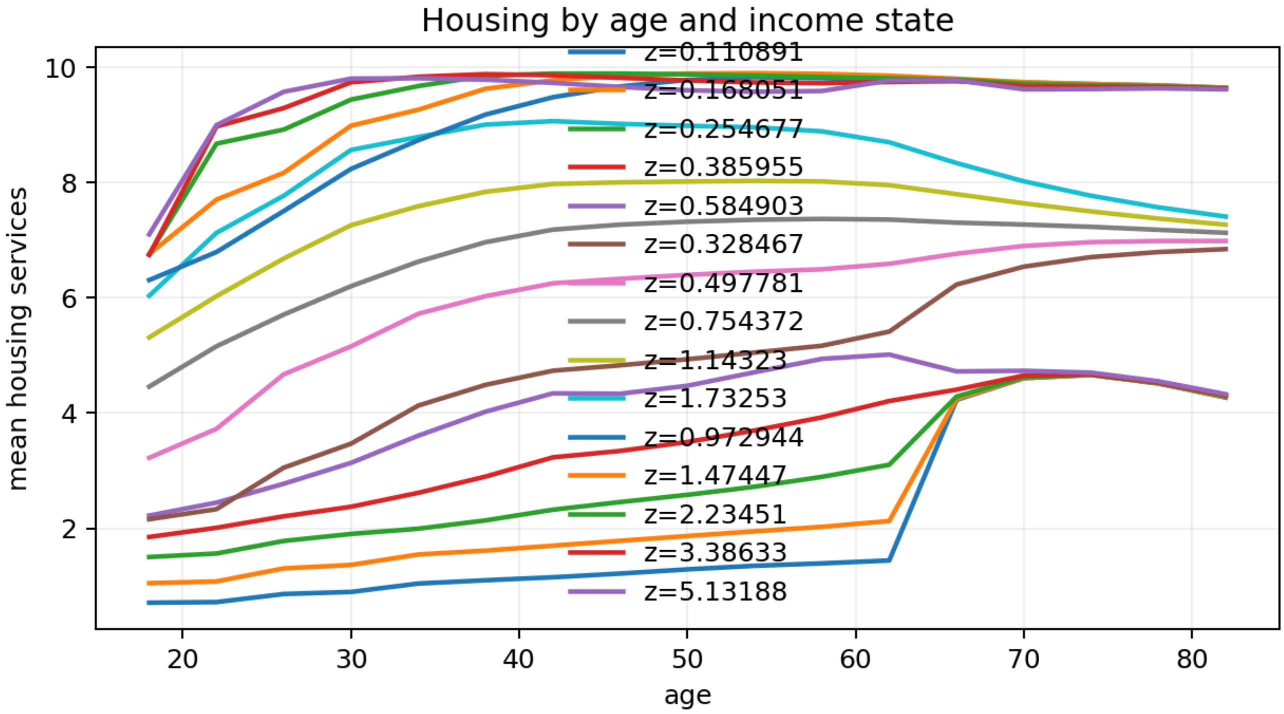
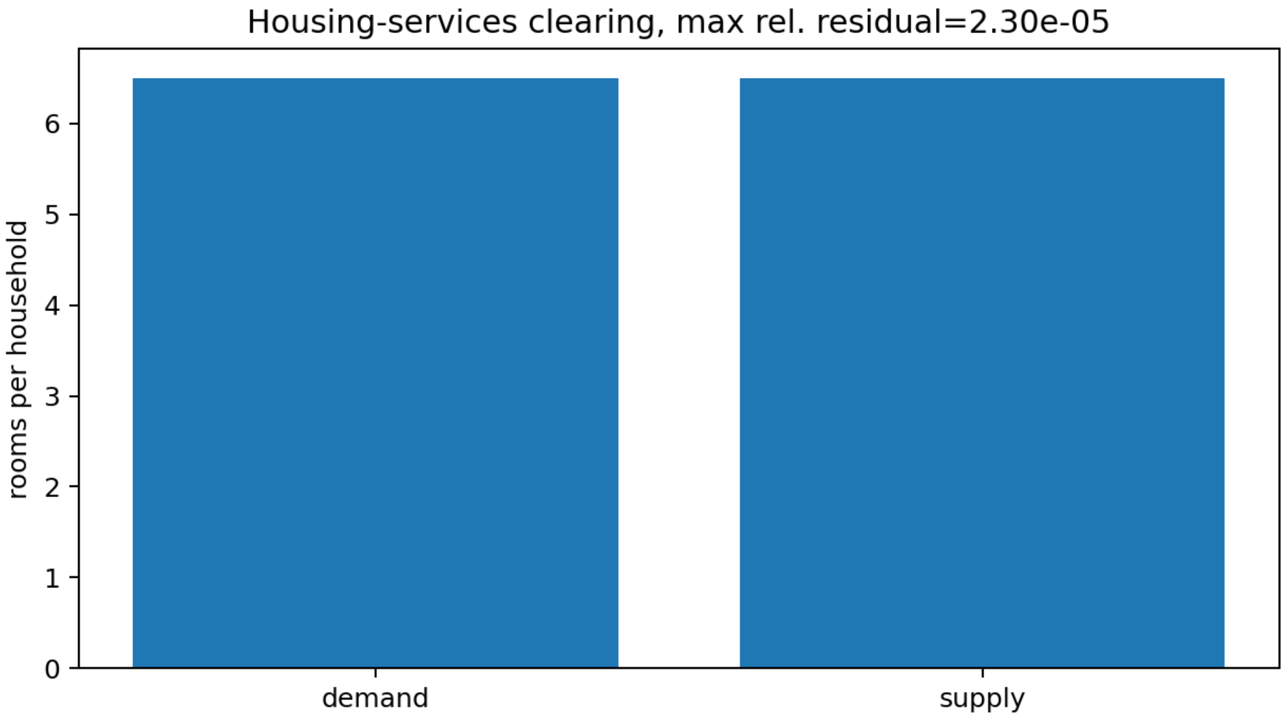
housing_by_age_income_state



housing_market



Standard initial-calibration diagnostics

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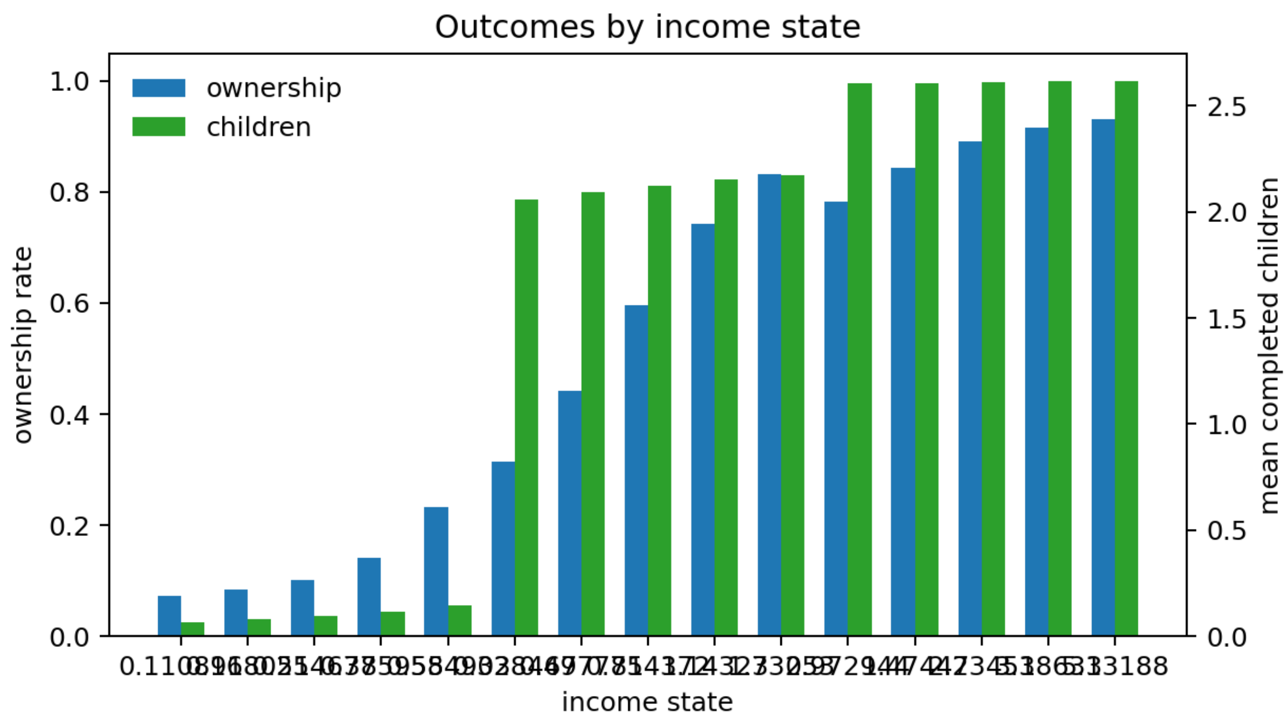
housing_prices

income_state_outcomes



Standard initial-calibration diagnostics

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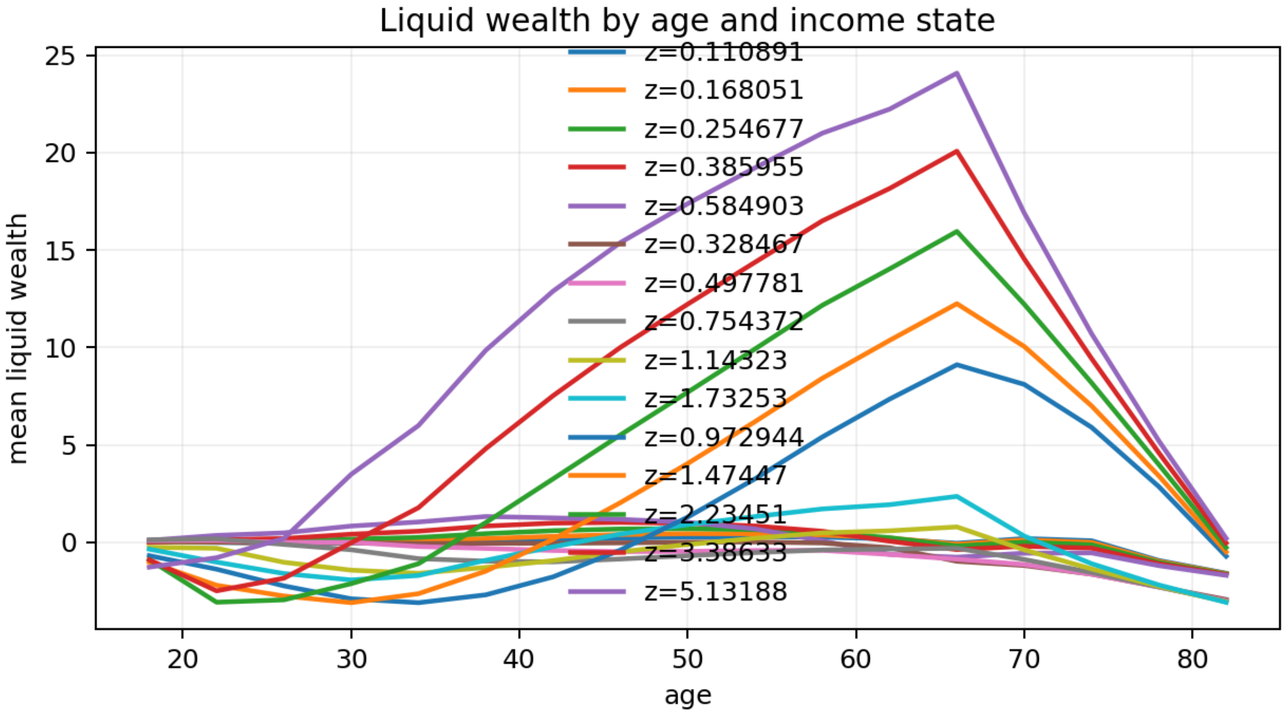
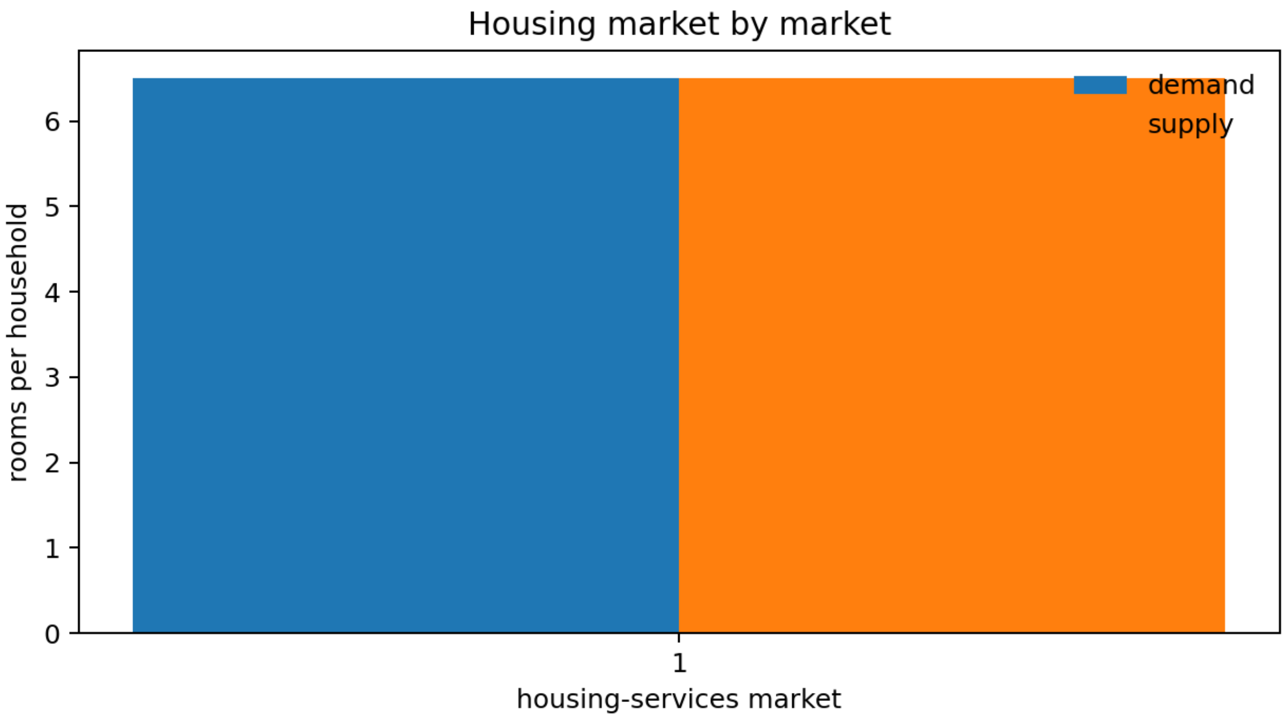
liquid_wealth_by_age_income_state



market_clearing_by_market



Standard initial-calibration diagnostics

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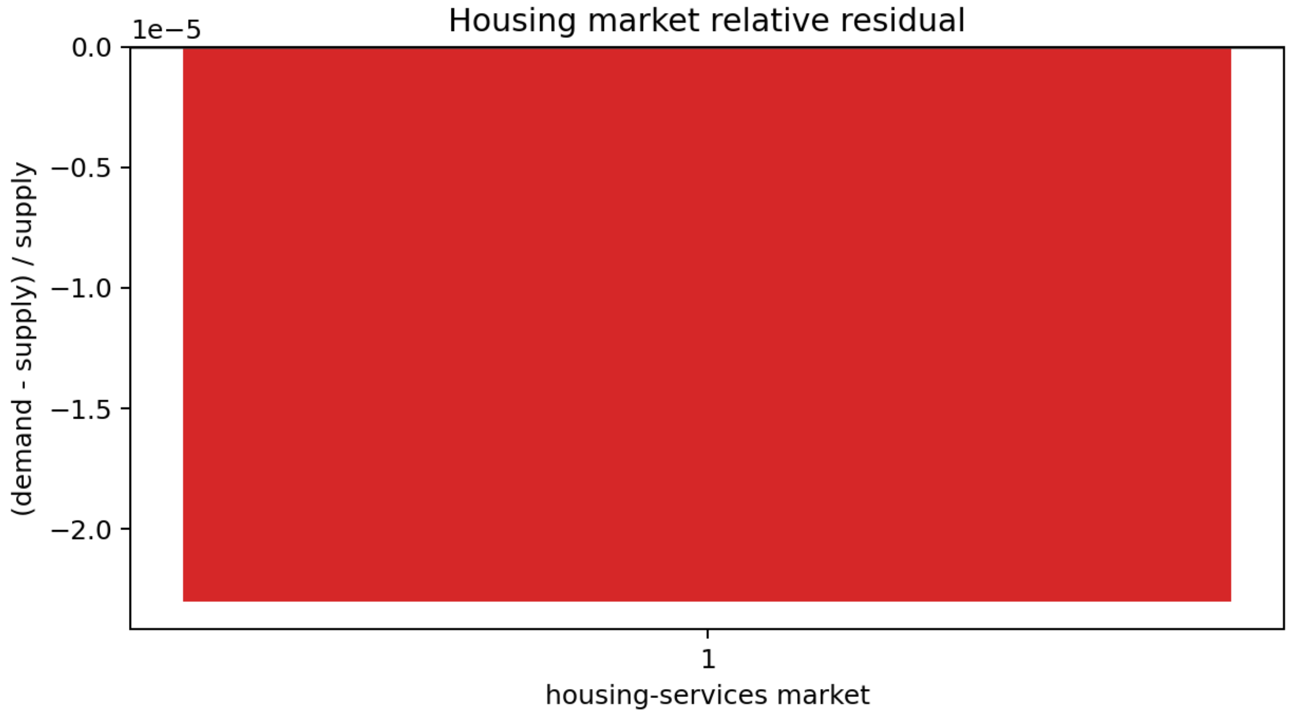
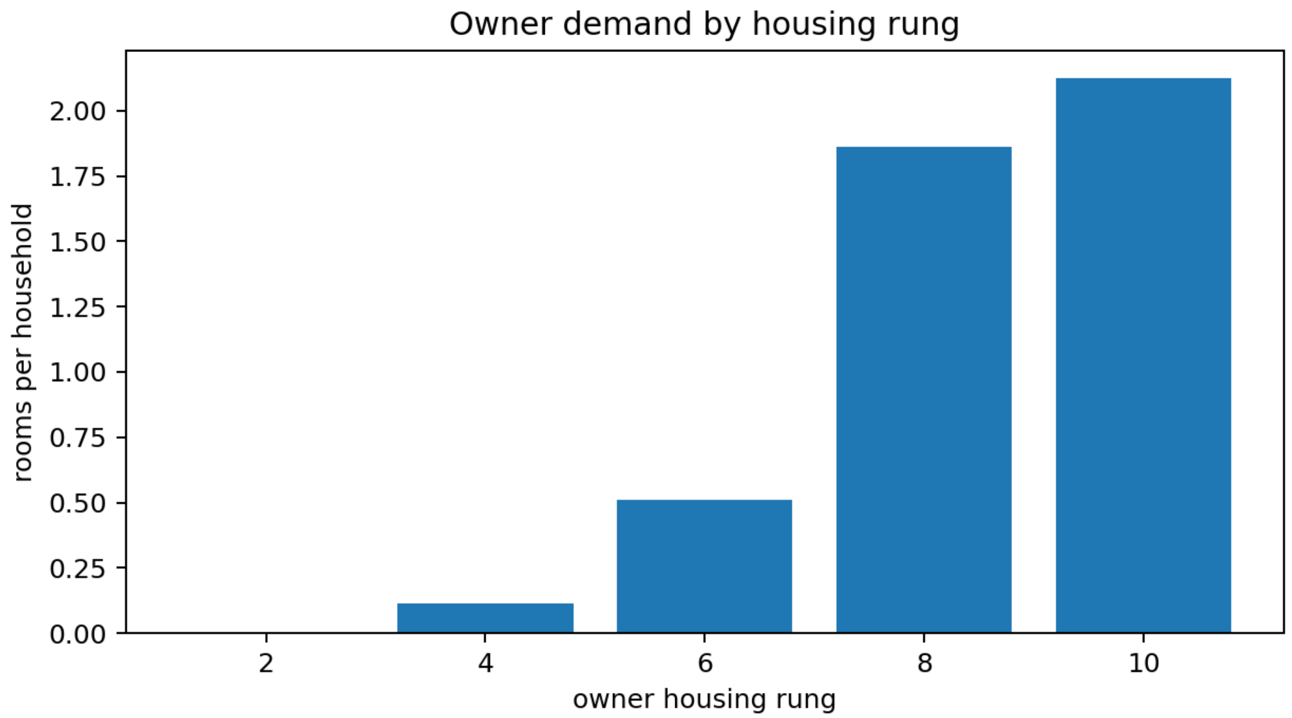
market_clearing_residuals



owner_rungs



Standard initial-calibration diagnostics

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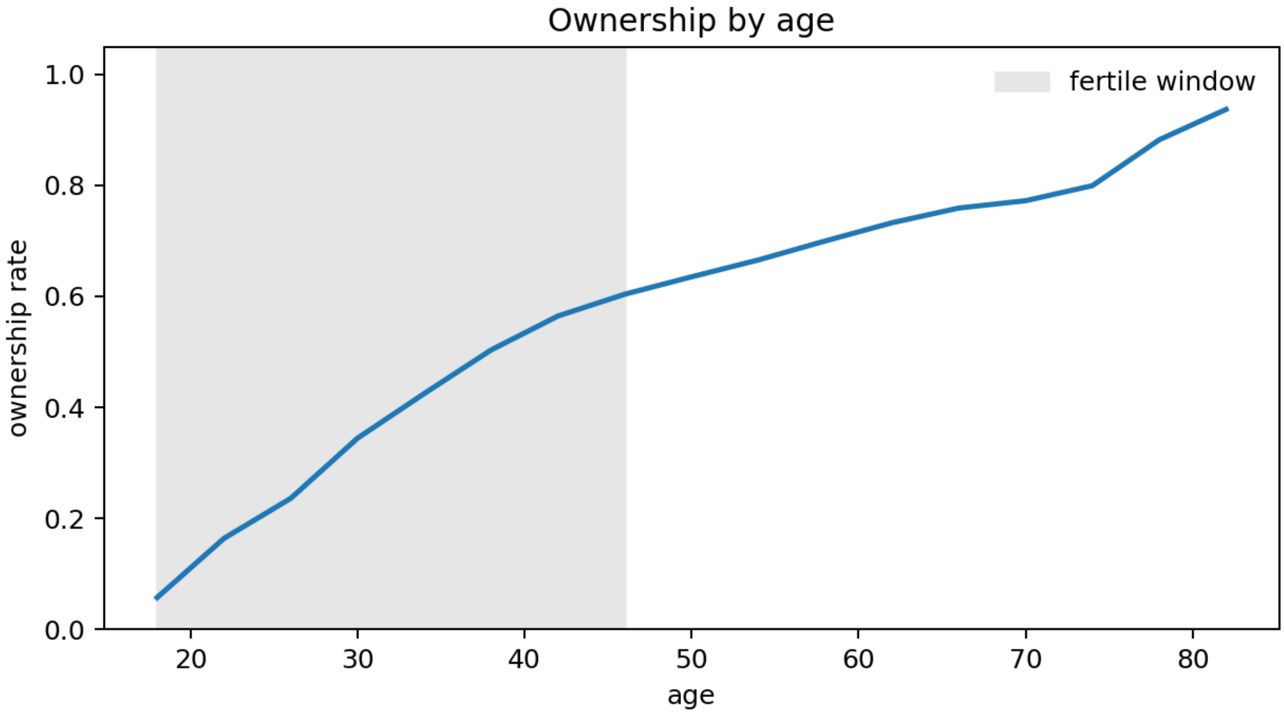
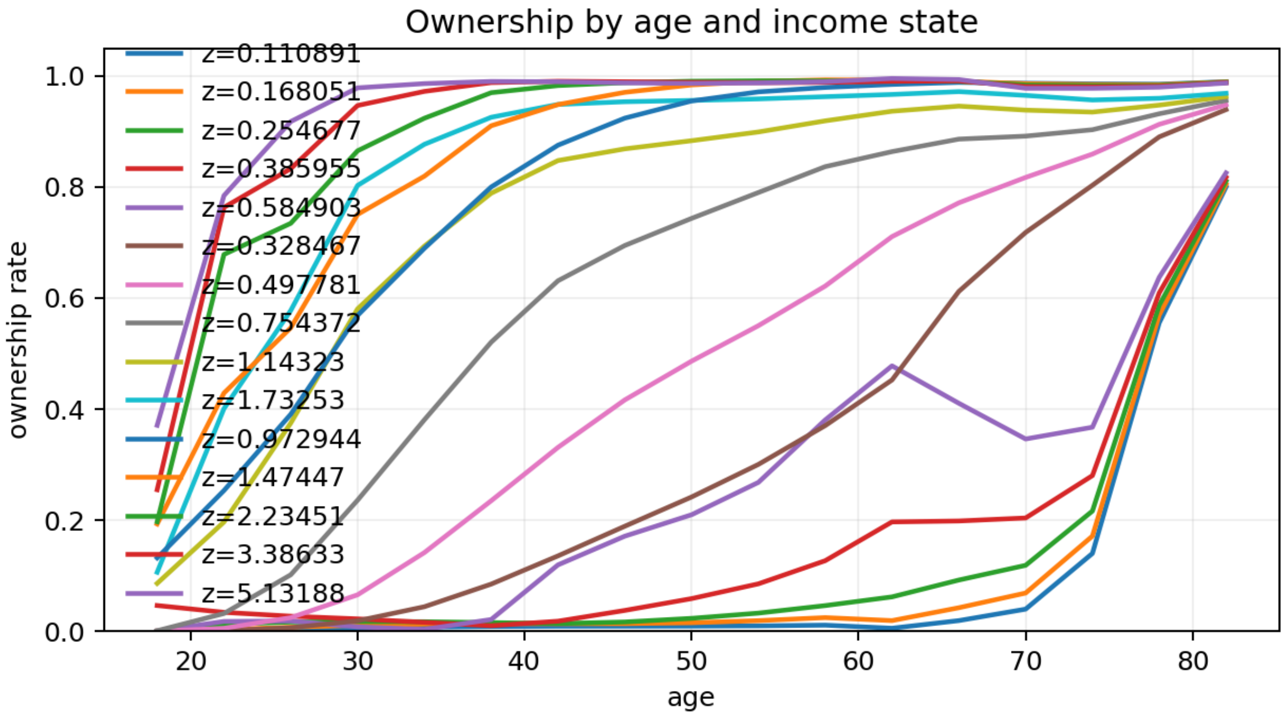
ownership_by_age



ownership_by_age_income_state



Standard initial-calibration diagnostics

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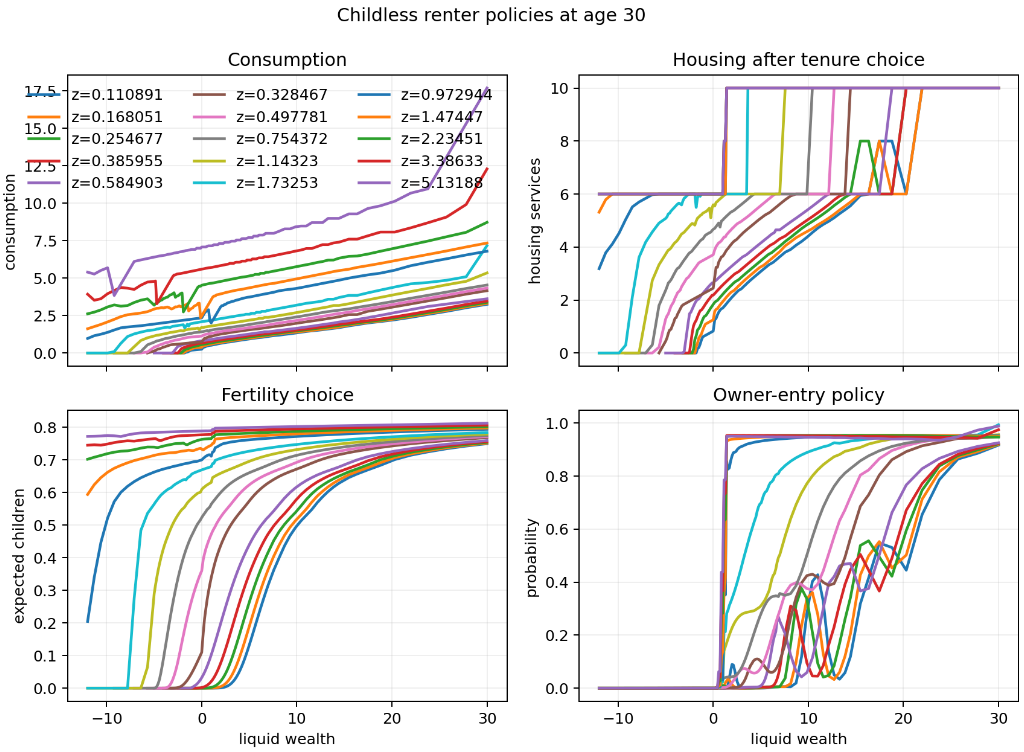
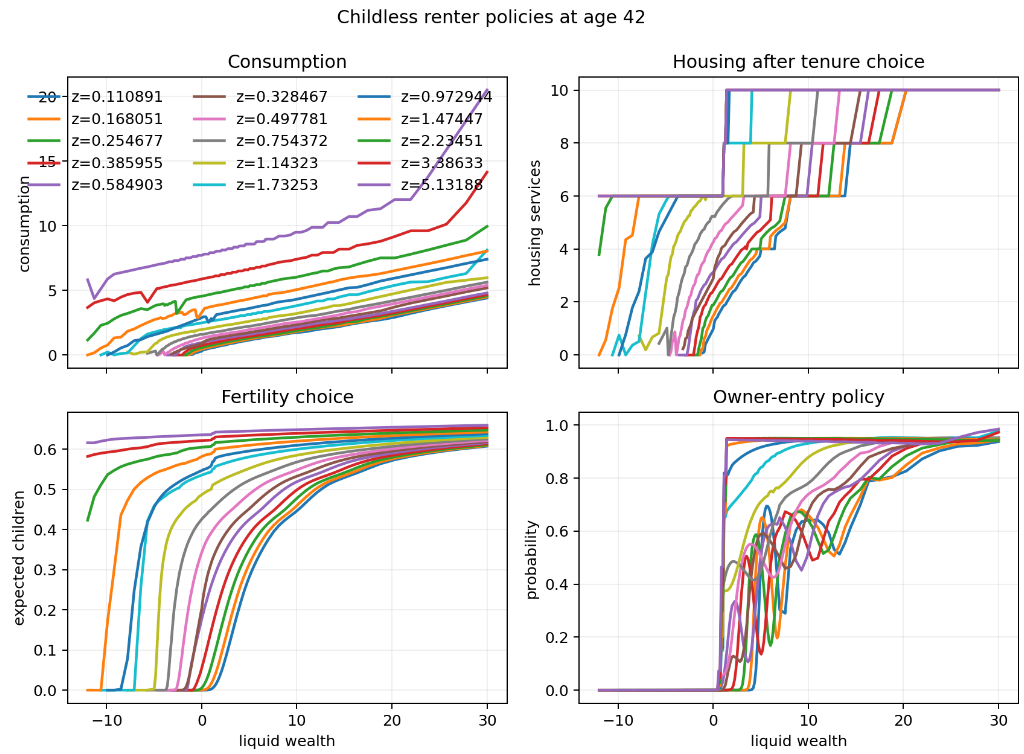
policy_childless_renter_age30



policy_childless_renter_age42



Standard initial-calibration diagnostics

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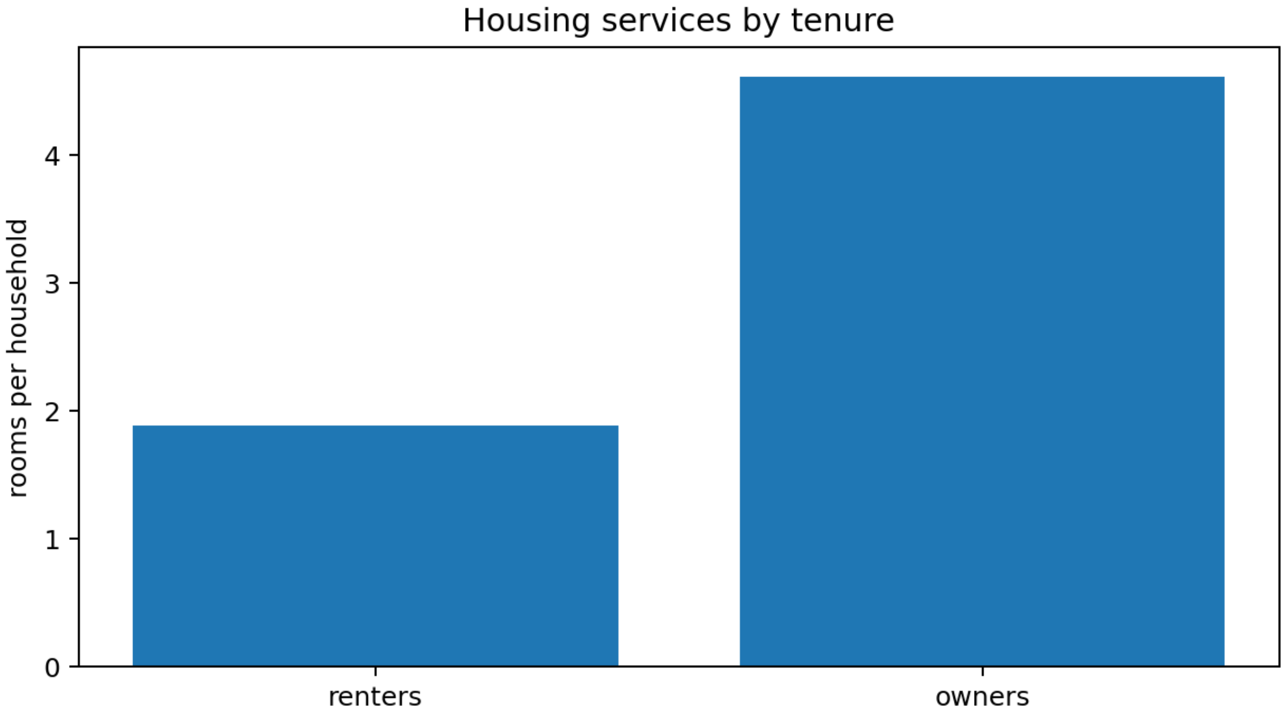
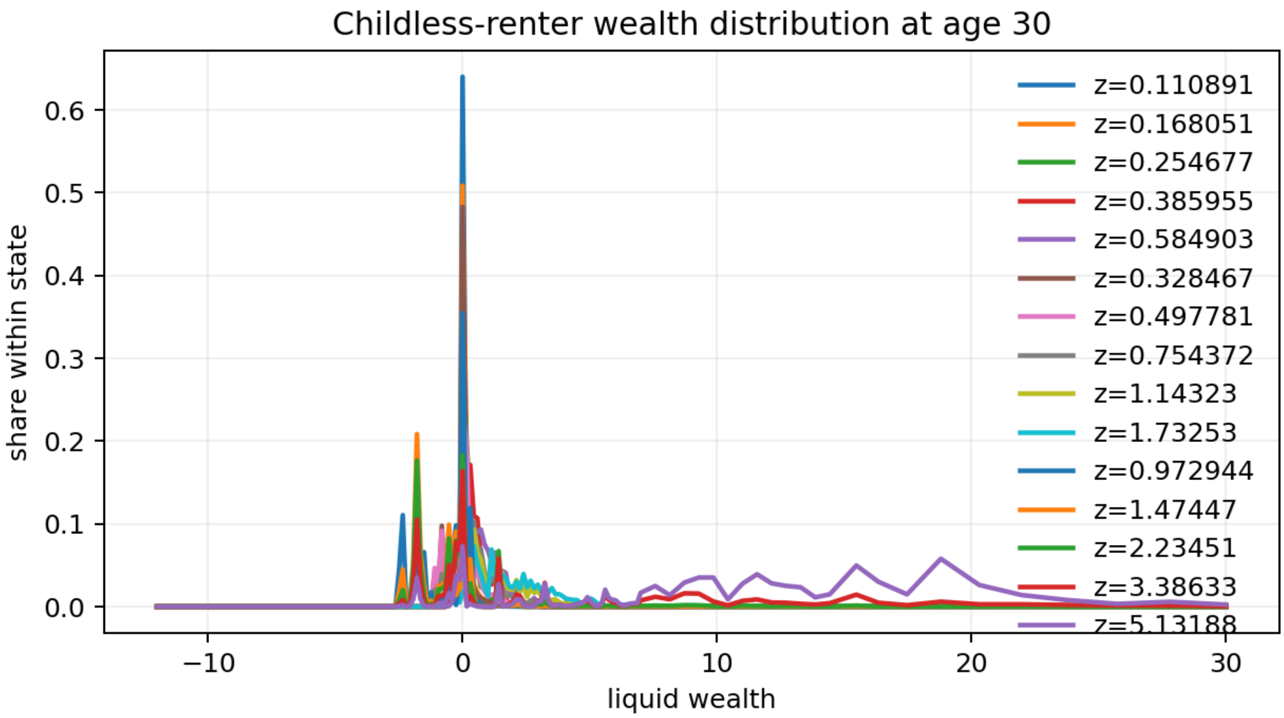
tenure_services



wealth_dist_childless_renter_age30



Standard initial-calibration diagnostics

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wealth_dist_childless_renter_age42